CHAPTER 3

Complex numbers and trigonometry

Addition formulas, exponential form, de Moivre's and Euler's formulas

Objectives. Establish the addition and double-angle formulas; write a complex number in exponential form and choose the form suited to a computation; compute powers; use de Moivre's and Euler's formulas to transform trigonometric expressions.

Prerequisites. Chapter 2; scalar product (Première); algebraic properties of the exponential function; unit circle and associated angles.

Fluency

Fluency · No calculator, five minutes.

1. Give $\cos\frac{\pi}{6}$ and $\sin\frac{5\pi}{6}$.
2. Express $\cos(\pi - x)$ and $\sin\left(\frac{\pi}{2} - x\right)$ in terms of $\cos x$ or $\sin x$.
3. Recall two expressions of the scalar product $\vec{a} \cdot \vec{b}$.
4. Simplify $e^a \times e^b$ and $(e^a)^n$.
5. Solve in $(-\pi, \pi]$ the equation $\cos x = \frac{1}{2}$.
6. Give the modulus and an argument of $-1 - i$.

Introductory activity

Activity 1 · A function that turns sums into products 35 min. For every real number θ, let $f(\theta) = \cos\theta + i\sin\theta$.

1. Compute $f(0)$, $f\left(\frac{\pi}{2}\right)$, $f(\pi)$. What is the modulus of $f(\theta)$? Where does the image of $f(\theta)$ lie?
2. Write $f\left(\frac{\pi}{3}\right)$ and $f\left(\frac{\pi}{6}\right)$ in algebraic form, then compute the product $f\left(\frac{\pi}{3}\right) f\left(\frac{\pi}{6}\right)$. Compare it with $f\left(\frac{\pi}{2}\right)$.
3. Compute likewise $f\left(\frac{\pi}{4}\right)^2$ and compare it with $f\left(\frac{\pi}{2}\right)$.
4. With a calculator or a spreadsheet, compare $f(a)f(b)$ and $f(a+b)$ for three pairs (a,b) chosen at random.
5. Which known function satisfies a relation of the same kind, $g(a)g(b) = g(a+b)$ and $g(0) = 1$?
6. Expand $f(a)f(b)$. Which trigonometric formulas would one need to know to prove the conjecture?

Course notes

Addition and double-angle formulas

Property · Addition formulas. For all real numbers a and b:

$$\cos(a-b) = \cos a \cos b + \sin a \sin b, \qquad \cos(a+b) = \cos a \cos b - \sin a \sin b,$$

$$\sin(a-b) = \sin a \cos b - \cos a \sin b, \qquad \sin(a+b) = \sin a \cos b + \cos a \sin b.$$

Proof. In a direct orthonormal frame, let $\overrightarrow{p}\begin{pmatrix}\cos a\\ \sin a\end{pmatrix}$ and $\overrightarrow{q}\begin{pmatrix}\cos b\\ \sin b\end{pmatrix}$: these vectors are unit vectors and $(\overrightarrow{u}, \overrightarrow{p}) \equiv a$, $(\overrightarrow{u}, \overrightarrow{q}) \equiv b \pmod{2\pi}$. By Chasles' relation, $(\overrightarrow{q}, \overrightarrow{p}) \equiv a - b \pmod{2\pi}$. We compute $\overrightarrow{q} \cdot \overrightarrow{p}$ in two ways:

$$\overrightarrow{q} \cdot \overrightarrow{p} = \|\overrightarrow{q}\| \, \|\overrightarrow{p}\| \cos(a-b) = \cos(a-b) \quad \text{and} \quad \overrightarrow{q} \cdot \overrightarrow{p} = \cos b \cos a + \sin b \sin a.$$

This proves the first formula. Replacing b by $-b$ and using $\cos(-b) = \cos b$, $\sin(-b) = -\sin b$, we obtain $\cos(a+b)$. Finally $\sin(a+b) = \cos\left(\frac{\pi}{2} - a - b\right) = \cos\left(\left(\frac{\pi}{2} - a\right) - b\right)$, which the first formula expands into $\sin a \cos b + \cos a \sin b$; we deduce $\sin(a-b)$ by replacing b by $-b$. □

Property · Double-angle formulas. For every real number a: $\sin 2a = 2\sin a \cos a$ and $\cos 2a = \cos^2 a - \sin^2 a = 2\cos^2 a - 1 = 1 - 2\sin^2 a$.

Example 1 · An exact value. $\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

Product and arguments

Property · Product of two trigonometric forms. For all real numbers θ and θ': $(\cos\theta + i\sin\theta)(\cos\theta' + i\sin\theta') = \cos(\theta+\theta') + i\sin(\theta+\theta')$.

Consequently, for all nonzero complex numbers z, z' and every integer n:

$$\arg(zz') \equiv \arg z + \arg z', \qquad \arg\frac{1}{z} \equiv -\arg z,$$

$$\arg\frac{z}{z'} \equiv \arg z - \arg z', \qquad \arg(z^n) \equiv n \arg z \pmod{2\pi}.$$

Proof. Expanding gives $(\cos\theta\cos\theta' - \sin\theta\sin\theta') + i(\sin\theta\cos\theta' + \cos\theta\sin\theta')$, which the addition formulas simplify. If $z = r(\cos\theta + i\sin\theta)$ and $z' = r'(\cos\theta' + i\sin\theta')$, then $zz' = rr'(\cos(\theta+\theta') + i\sin(\theta+\theta'))$ with $rr' > 0$: this is a trigonometric form of zz'. The other relations follow: $z \times \frac{1}{z} = 1$ has argument 0, then $\frac{z}{z'} = z \times \frac{1}{z'}$, and an induction for $n \geq 0$, extended to $n < 0$ through the inverse. □

Key point · Interpretation. Multiplying by $\cos\theta + i\sin\theta$ turns through an angle θ about O. The conjecture of the activity of chapter 2 is proved: multiplying by i is a direct quarter turn, multiplying by $1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ is an eighth of a turn followed by an enlargement with ratio $\sqrt{2}$.

Exponential form

Definition · Imaginary exponential. For every real number θ, we write $e^{i\theta} = \cos\theta + i\sin\theta$. Every nonzero complex number can be written $z = re^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg z \pmod{2\pi}$: this is its *exponential form*.

Property · Rules of computation. For all real numbers θ, θ' and every integer n:

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}, \qquad \frac{1}{e^{i\theta}} = e^{-i\theta} = \overline{e^{i\theta}}, \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}, \qquad \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Values to know: $e^{i0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{-i\pi/2} = -i$, $\left|e^{i\theta}\right| = 1$.

Property · De Moivre's formula. For every real number θ and every integer n: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Property · Euler's formulas. For every real number θ: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$.

Proof. $e^{i\theta} + e^{-i\theta} = e^{i\theta} + \overline{e^{i\theta}} = 2\,\mathrm{Re}\left(e^{i\theta}\right) = 2\cos\theta$, and $e^{i\theta} - e^{-i\theta} = 2i\,\mathrm{Im}\left(e^{i\theta}\right) = 2i\sin\theta$. □

Method · Choosing the suitable form. For a sum, the algebraic form; for a product, a quotient or a power, the exponential form. Example: $1 + i\sqrt{3} = 2e^{i\pi/3}$, so

$$\left(1 + i\sqrt{3}\right)^{10} = 2^{10}e^{10i\pi/3} = 1{,}024\,e^{-2i\pi/3} = -512 - 512\sqrt{3}\,i,$$

since $\frac{10\pi}{3} - 4\pi = -\frac{2\pi}{3}$.

Method · Expressing $\cos 3\theta$ in terms of $\cos\theta$: de Moivre. Expand $(\cos\theta + i\sin\theta)^3$ with the binomial formula and identify the real part: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$.

Method · Linearising $\cos^3\theta$: Euler. $\cos^3\theta = \dfrac{\left(e^{i\theta} + e^{-i\theta}\right)^3}{8} = \dfrac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \dfrac{\cos 3\theta + 3\cos\theta}{4}$.

Method · Half angle. To factorise a sum of exponentials, take out the exponential of the mean angle: $e^{ia} + e^{ib} = e^{i\frac{a+b}{2}}\left(e^{i\frac{a-b}{2}} + e^{-i\frac{a-b}{2}}\right) = 2\cos\frac{a-b}{2}\,e^{i\frac{a+b}{2}}$.

Watch out. $e^{i\theta}$ is not a positive real number: its modulus is 1, but it can be negative ($e^{i\pi} = -1$) or non-real. An expression $re^{i\theta}$ is an exponential form only if $r > 0$.

Exercises

Exercise 3.1 · Exponential forms. Write in exponential form: $-2+2i$; $-1-i\sqrt{3}$; $5i$; $\dfrac{1+i}{1-i}$.

Exercise 3.2 · Two ways of writing the same product. Let $z_1 = \sqrt{3}+i$ and $z_2 = 1-i$.

1. Write z_1, z_2 and then $z_1 z_2$ in exponential form.
2. Write $z_1 z_2$ in algebraic form.
3. Deduce the exact values of $\cos\frac{\pi}{12}$ and $\sin\frac{\pi}{12}$.

Exercise 3.3 · Powers.

1. Compute $(1-i)^8$ and $\left(\sqrt{3}+i\right)^{12}$.
2. Determine the natural numbers $n \geq 1$ for which $\left(\sqrt{3}+i\right)^n$ is real; positive real; purely imaginary. Give the smallest in each case.

Exercise 3.4 · De Moivre's formula. Using de Moivre's formula and the binomial formula, express $\cos 4\theta$ in terms of $\cos\theta$ only, and $\sin 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Exercise 3.5 · Linearising in order to integrate.

1. Using Euler's formulas, prove that $\sin^3 x = \dfrac{3\sin x - \sin 3x}{4}$.
2. Deduce the value of $\displaystyle\int_0^{\pi} \sin^3 x \,\mathrm{d}x$.

Exercise 3.6 · Modulus and argument of $1+\mathrm{e}^{i\theta}$.

1. Prove that for every real number θ, $1+\mathrm{e}^{i\theta} = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$.
2. Deduce the modulus and an argument of $1+\mathrm{e}^{i\theta}$ when $\theta \in (-\pi, \pi)$.
3. Application: modulus and argument of $1+\mathrm{e}^{2i\pi/3}$, then a check with the algebraic form.

Exercise 3.7 · A sum of cosines. Let θ be a real number that is not a multiple of 2π, and n a natural number.

1. Prove that $\displaystyle\sum_{k=0}^{n} \mathrm{e}^{ik\theta} = \frac{\mathrm{e}^{i(n+1)\theta}-1}{\mathrm{e}^{i\theta}-1}$.
2. Using the half-angle method, prove that this sum equals $\mathrm{e}^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$.
3. Deduce $\displaystyle\sum_{k=0}^{n} \cos(k\theta) = \frac{\cos\frac{n\theta}{2}\,\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, then check the formula for $n=2$ and $\theta=\frac{\pi}{2}$.

Problem

Problem · Superposing two oscillations. In physics, a simple voltage or sound can be written $t \mapsto A\cos(\omega t + \varphi)$. We show here that a sum $a\cos x + b\sin x$ can be put in this form. We study $g(x) = \sqrt{3}\cos x + \sin x$.

1. Write $\sqrt{3} - i$ in exponential form.
2. Prove that $g(x) = \mathrm{Re}\left(\left(\sqrt{3} - i\right)\mathrm{e}^{ix}\right)$, then that $g(x) = 2\cos\left(x - \frac{\pi}{6}\right)$.
3. Deduce the maximum and the minimum of g on $\mathbb{R}$, and the real numbers in $[0, 2\pi)$ where they are reached.
4. Solve in $[0, 2\pi)$ the equation $g(x) = \sqrt{2}$.
5. Generalise: let a and b be two real numbers, not both zero. Prove that there exist $R > 0$ and a real number φ such that, for every real number x, $a\cos x + b\sin x = R\cos(x - \varphi)$. Express R in terms of a and b.